Chapter 7: Nonparametric and Categorical Data

Statistics for Environmental Science

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Parametric vs. nonparametric tests

Chapter 6 covered three forms of the t-test (one-sample, paired, and independent-samples, also called two-sample). All three need a continuous outcome that is roughly normal (for the paired test, the difference scores), and the pooled independent-samples test also assumes similar variance across the two groups. This chapter covers what to do when those assumptions aren't met. We'll also cover what to do if your data aren't continuous at all.

Every test needs a known null distribution for its test statistic (Chapter 4), because that is the shape we compare our result against to get a p-value. A **parametric** test is one whose null distribution we can know only by assuming something specific about the shape of the population the data came from. Every test so far in this book has been parametric.

For the t-test, that assumption is normality. The t-distribution isn't the shape of the raw data. It's the shape the t-statistic takes across repeated samples, and we can only derive it because we assumed the population itself was normal, or relied on the central limit theorem (Chapter 3) for a large enough sample.

When the distributional assumptions are met, parametric tests generally have the most power (Chapter 5) of any test available. However, when distributional assumptions are not met, a parametric test's p-value can be too small or too large, and without simulation it is hard to tell which. The worse the departure from the assumed distribution, the larger the possible error.

A **nonparametric test** is a test whose null distribution does not depend on the population having a particular shape, such as normal. These tests still make assumptions, and those assumptions are usually easier to meet than normality.

The **rank** of a value is its position when the data are put in order from smallest to largest (first, second, third, and so on). Many nonparametric tests are run on the ranks instead of the raw values. The median is a familiar statistic that depends only on this ordering. It is the middle value once the data are put in order.

Rank-based tests also work on ordinal data. Because the rank-based tests in this chapter only need a rank ordering, they are not limited to interval- or ratio-scale measurements (Chapter 1) the way a t-test is. A t-test needs a mean, which requires numbers where the *distance* between values means something. Plenty of environmental data doesn't work that way: a water-quality rating (poor, fair, good, excellent), a disturbance-severity class, a Likert-style survey response. You can put those in order, but "the average disturbance class was 2.4" isn't meaningful the way "the average pH was 6.8" is.

This chapter covers two different kinds of tools:

- **Nonparametric alternatives (to t-tests).** Your outcome is still continuous, but it's skewed, has outliers, or your sample is too small to trust normality. The **Wilcoxon signed-rank test** (Wilcoxon 1945) answers the same question as the one-sample or paired t-test without assuming normality. The **Mann-Whitney-Wilcoxon test** (Mann and Whitney 1947) answers the same question as the independent-samples t-test without assuming normality.
- **Categorical data.** Your outcome is a category (Chapter 1) instead of a number. A site either has an invasive species or it doesn't, and a sample either exceeds a threshold or it doesn't. None of the tests above apply here. The **chi-square (χ^2) tests** compare the counts observed in each category with the counts expected under H_0.

We'll state the assumptions explicitly for each test as we go through them.

The Wilcoxon signed-rank test

Chapter 6 built the paired t-test around this same example, wetland restoration and soil moisture, with a full 20-plot dataset, and drew the design in Figure 6.8. The same plots are measured before and after, so each plot contributes one difference. Let's run the same design on a separate, smaller study of 12 plots, and this time check whether we should trust the result before reporting it. The data in this example are invented. The section runs the paired t-test, checks its assumption, and then runs two nonparametric tests on the same plots, the Wilcoxon signed-rank test and the sign test.

Table 1: Soil moisture (%) before and after restoration, 12 plots.

Plot	Before	After	Difference
1	20	26	6
2	15	21	6
3	22	21	-1
4	18	22	4
5	16	23	7
6	24	23	-1

Plot	Before	After	Difference
7	19	18	-1
8	14	22	8
9	21	21	0
10	17	17	0
11	25	25	0
12	20	23	3

Step 1: run the paired t-test, the way Chapter 6 taught it.

Table 2: Paired t-test: soil moisture before vs. after restoration.

Mean difference	t	df	p
2.58	2.57	11	0.026

$t(11) = 2.57$, $p = 0.026$, a significant result. Restoration looks like it increased soil moisture.

Step 2: check the assumptions. Let's look at the differences before trusting that p-value.

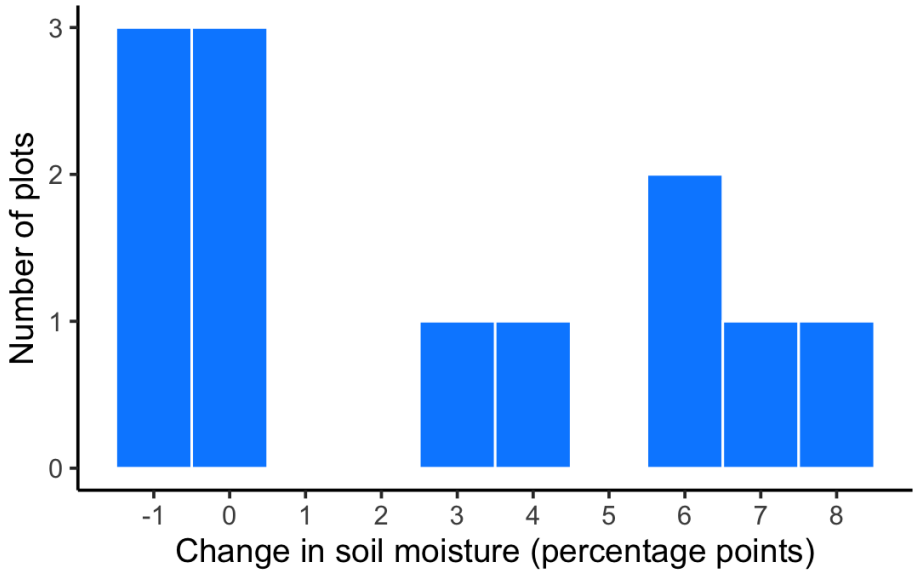


Figure 1: Change in soil moisture per plot. Three plots showed no change, and the positive changes stretch much further than the negative ones. A paired t-test assumes a symmetric, bell-shaped pattern.

Table 3: Shapiro-Wilk normality test on the differences.

Statistic	p
0.85	0.039

The Shapiro-Wilk test (Chapter 6) has the null hypothesis that the data came from a normal distribution. Here $p = 0.039$, below 0.05, so the test rejects normality for the differences and agrees with the picture.

Three plots didn't change at all, three dropped slightly, and the rest rose by 3 to 8 points. That is not symmetric, let alone normal. The t-test's "significant" result depends on an assumption the data don't meet, so we turn to the Wilcoxon signed-rank test.

The question. Did soil moisture systematically change after restoration, or is any change we see just plot-to-plot noise?

- H_0: the median difference is 0, so positive and negative ranks balance.
- H_a: the median difference is not 0.

How it works, conceptually. Instead of using the actual sizes of the differences, the test ranks the *absolute* differences from smallest to largest, then checks whether the positive-signed ranks and negative-signed ranks are roughly balanced. Balanced ranks are the null case, meaning a plot's rank (how extreme its change was) has nothing to do with whether that change was an increase or a decrease. If soil moisture really increased at most plots, the larger ranks should belong mostly to positive differences instead. Because the test uses ranks, the plot that rose by 8 points gets the top rank, 9, and it would get the same rank if it had risen by 30. The top rank is 9 instead of 12 because the three plots with no change drop out of the test, as explained below. One extreme plot can't dominate the result the way it could dominate a mean.

Its assumption. The Wilcoxon signed-rank test does not require normality, but it assumes the differences come from a symmetric distribution around their median. That's a milder assumption than normality (symmetric-but-not-normal distributions are common), but our histogram above strains it: the positive side stretches to 8, the negative side barely reaches -1. We'll run the test anyway, because its symmetry assumption is still milder than the t-test's normality assumption and the 8-point plot counts only as the top rank. We will note the problem when we interpret the result.

```
wilcox.test(differences, mu = 0)
```

Table 4: Wilcoxon signed-rank test on restoration soil-moisture differences.

Statistic (V)	p
39	0.057

The three plots with no change drop out of the test. A zero difference can't be ranked as positive or negative, so the signed-rank test runs on the remaining 9 plots instead of all 12. The rank test already ignores the exact size of each change and keeps only its order, and dropping the zeros means it also works with less data than the t-test. The zeros and the tied differences (two plots at $+6$, three at -1) also keep R from using the exact null distribution, so R uses a normal approximation instead. The exact null distribution is built below, after we calculate V. The normal approximation replaces it with a normal curve that has the same mean and SD.

How V is calculated. Drop the zero differences, take the absolute value of what's left, rank those absolute values from smallest to largest (tied absolute values share the average of their ranks), then sum the ranks belonging to the *positive* differences. For example, the three differences of -1 all have absolute value 1 and would take ranks 1, 2 and 3, so each gets rank 2. The two differences of $+6$ would take ranks 6 and 7, so each gets 6.5. That sum is V.

Table 5: Ranking the nonzero restoration differences to calculate V.

Difference	Abs. difference	Rank	Sign
6	6	6.5	+
6	6	6.5	+

Difference	Abs. difference	Rank	Sign
-1	1	2.0	-
4	4	5.0	+
7	7	8.0	+
-1	1	2.0	-
-1	1	2.0	-
8	8	9.0	+
3	3	4.0	+

V is the sum of the ranks where Sign is "+": $6.5 + 6.5 + 5 + 8 + 9 + 4 = 39$, matching the table R gave us above.

How the null distribution of V is built. Under H_0, each plot's rank has an independent 50/50 chance of being positive or negative. With $n = 9$ nonzero differences (not 12, since the zeros dropped out), that gives $2^9 = 512$ equally likely sign patterns. No new plots are measured. Each pattern gives one V, the value our data would have produced if each plot's change had been equally likely to go up or down. Listing the V for all 512 patterns gives the exact null distribution of V, shown in Figure 2.

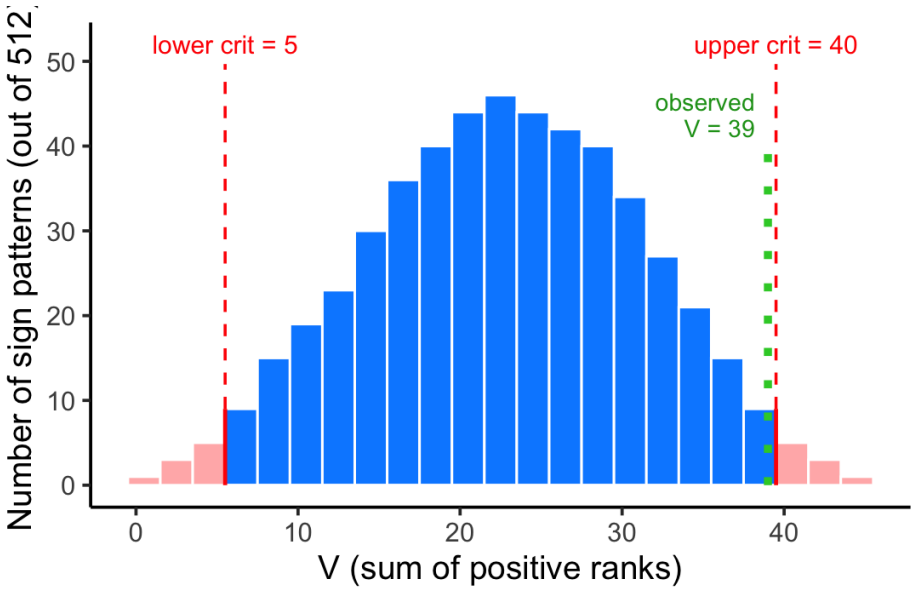


Figure 2: Exact null distribution of the Wilcoxon signed-rank statistic V, from all 512 equally likely sign patterns of the 9 nonzero ranked differences. Shaded regions mark the two-tailed 5% rejection region. Our observed V = 39 falls just short of the upper critical value, 40, outside the rejection region.

Our observed $V = 39$ falls just short of the upper critical value, 40, so it is not in the 5% region. V can only take certain values, so the boundary can't fall exactly at 2.5%. A V of 39 or more comes from just over 2.5% of the 512 patterns, which puts the exact two-tailed p at 0.051. The result is not significant, though it is close to the cutoff. With zeros and tied differences, R uses a normal approximation to this distribution instead of the exact one, so its p-value, 0.057, differs slightly from the exact figure.

The test does not need normality because nothing in this distribution assumed a shape for soil moisture. The only inputs were our 9 ranks and the 50/50 sign rule from H_0.

So far. t-test $p = 0.026$ (significant), signed-rank test $p = 0.057$ (not significant). The two tests reach different verdicts on the same 12 plots because the t-test's numerator and denominator both used the exact magnitudes the signed-rank test ignores. In the t-test, the increases of 3 to 8 points count at their full size.

The sign test. The **sign test** looks only at whether each difference is positive or negative. It is an even simpler nonparametric option than the signed-rank test, and it asks whether pluses outnumber minuses more than chance would suggest. The sign test makes no assumption about the distribution of the differences, not even symmetry. Its only assumption is the one every test in this book shares, independent observations from a sound study design (see Chapter 6).

We already flagged the symmetry assumption as shaky for these data, so the sign test avoids the one assumption the signed-rank test still needed. Assuming less costs power, though, because the sign test throws away the size of every difference, so it is usually a last resort.

The sign test uses the binomial distribution from Chapter 3. `binom.test(6, 9, p = 0.5)` asks how often chance alone would give 6 or more increases out of 9, or 3 or fewer (the same distance from 4.5 in the other direction, because the test is two-tailed), if each plot were equally likely to go up or go down ($\theta = 0.5$, the binomial success probability).

```
binom.test(6, 9, p = 0.5)
```

Table 6: Sign test on restoration soil-moisture differences.

Plots increased	Plots total	p
6	9	0.508

Six of the 9 nonzero plots increased, and $p = 0.508$ is far from significant.

The full picture. The three tests reach different verdicts on the same 12 plots, and each uses less of the data than the one before it (Table 7).

Table 7: Three tests on the same 12 restoration plots.

	Paired t-test	Wilcoxon signed-rank	Sign test
What it uses	Full size of each difference	Ranks of the nonzero differences	Direction only
Plots used	12	9	9
Assumption about shape	Differences roughly normal	Differences symmetric around their median	None
Result	$t(11) = 2.57$	$V = 39$	6 of 9 increased
p	0.026	0.057	0.508

The t-test gave the smallest p-value here because it used the full size of every change, including the increases of 3 to 8 points. That p-value is trustworthy only if the differences are roughly normal, and these are not. Check the assumptions before you run the test, so the result cannot influence which test you choose. The examples in this chapter ran the t-test first only to show how far its p-value can move when an assumption fails.

Table 8 collects the Wilcoxon signed-rank test on one card.

Table 8: Test card: Wilcoxon signed-rank test.

Purpose	Did the units change, on average? Paired data whose differences are not normal
Hypotheses	H_0: the median difference is 0 (positive and negative ranks balance). H_a: it is not
Assumptions	Differences symmetric about their median (milder than normal)
Test statistic	V = sum of the ranks of the positive differences. Zeros drop out
Report	Direction, the median difference across all units, V, p, and the number of nonzero differences

The Mann-Whitney-Wilcoxon test

The same check applies to the independent-samples t-test. Many cities are replacing orange sodium streetlights with white LED lights, and ecologists want to know whether the switch changes how much bats use lit areas. Researchers placed acoustic detectors at 8 sites under sodium lights and 8 sites under LED lights and counted bat passes per night, the number of times a bat flew past the detector. The data in this example are invented. Figure 3 gives the design.

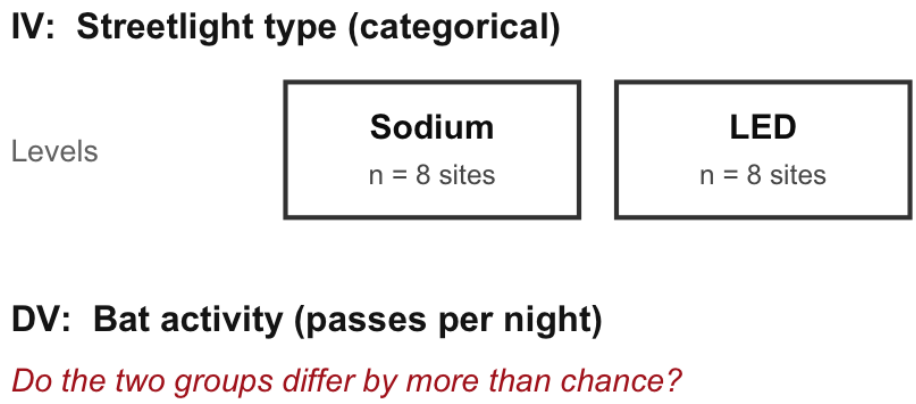


Figure 3: An independent-groups design. Unlike the paired design earlier in this chapter, the two boxes hold different sites, so there is nothing to subtract within a unit.

Table 9: Bat passes per night at sites under sodium and LED streetlights.

Site	Sodium	LED
1	21	17
2	16	18
3	22	18
4	16	12
5	17	11
6	16	11
7	17	11
8	34	10

Step 1: run the t-test, the way Chapter 6 taught it.

Table 10: Independent-samples t-test: sodium vs. LED bat activity.

Mean difference	t	df	p
6.38	2.54	14	0.024

$t(14) = 2.54$, $p = 0.024$, a significant result. Bat activity looks higher under sodium lights.

Step 2: check the assumptions. Chapter 6 said to check them before trusting a p-value, so we look at the data now.

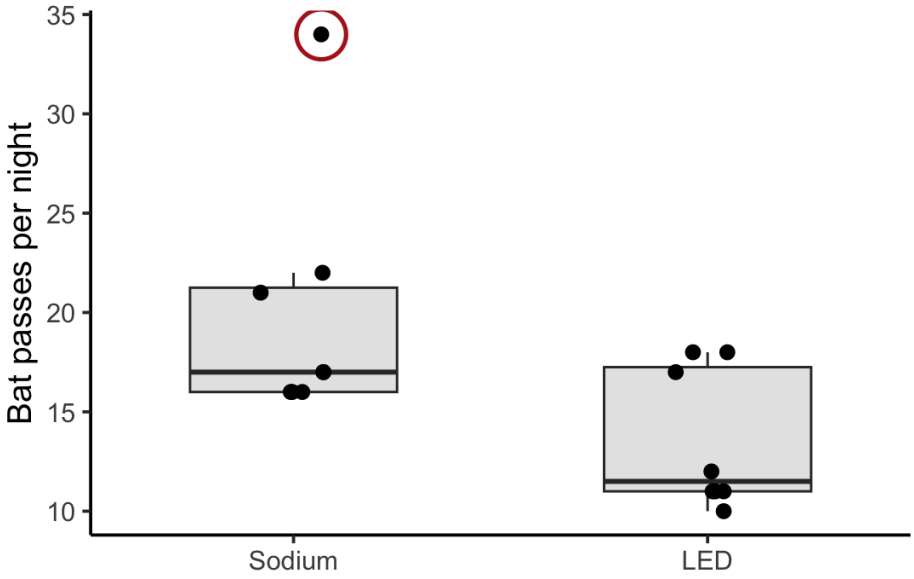


Figure 4: Bat passes per night by streetlight type. The circled sodium site has an unusually high count, visibly pulling that group's distribution away from a normal, bell-shaped one.

The circled site has 34 passes, 12 more than any other site in either group. We don't just throw away outliers, since the site may be near a hedgerow or tree line that bats follow at night. But it does make the sodium group far from normal:

Table 11: Shapiro-Wilk normality test, by group.

Group	Statistic	p
Sodium	0.70	0.002
LED	0.77	0.015

Both groups fail the Shapiro-Wilk test, sodium with $p = 0.002$ and LED with $p = 0.015$. The t-test's "significant" result depends on an assumption the data don't meet. The Mann-Whitney-Wilcoxon test is designed for this situation.

The question. Does bat activity differ between sites under sodium and LED lights, or is the difference we see consistent with random site-to-site variation?

- H_0: bat activity has the same distribution at sodium and LED sites, so ranks are well mixed between the groups.

- H_a: bat activity tends to rank higher at one type of site than the other.

How it works, conceptually. The signed-rank test ranked the absolute differences within one set of paired differences. The Mann-Whitney-Wilcoxon test ranks the raw values from both groups together, because independent groups have no pairs to subtract. Pool all the observations from both groups together, rank them from smallest to largest, then compare the sum of ranks in Group A to the sum of ranks in Group B. If the groups don't differ, their ranks should be well mixed. If one group tends to have higher values, its ranks should be higher too. The outlier site gets the highest rank, 16, whether it had 23 passes or 34. Its size enters the t-test through both the mean and the standard deviation, but the Mann-Whitney-Wilcoxon test uses only its rank.

Its assumption. The Mann-Whitney-Wilcoxon test assumes both groups' population distributions have the same shape. It does not require normality. Distributions with the same shape also have equal variances, but equal variances alone do not meet the assumption, because two distributions can share a variance and differ in shape. If the two groups have differently shaped distributions (say, one skewed and one symmetric), the test still runs. A significant result could then come from a difference in center (location), in shape, or both, so it no longer tells you only that the medians differ.

Our two groups don't quite share a shape. The sodium group is strongly right-skewed, mostly because of the one 34-pass site, while the LED group is only mildly skewed. So read a result from this test as "bat activity tends to be higher in one group," not strictly as a difference in medians.

```
wilcox.test(sodium, led)
```

Table 12: Mann-Whitney-Wilcoxon test: sodium vs. LED bat activity.

Statistic (U)	p
50	0.063

R labels this statistic W. It is the same number as the Mann-Whitney U, and this chapter calls it U. The pooled counts include ties (three sites each at 11, 16 and 17, and two at 18), so R again uses a normal approximation instead of the exact null distribution.

How U is calculated. Pool all 16 counts, rank them from smallest to largest (tied counts share the average of their ranks), then add up the ranks belonging to the sodium group. Subtract $n_A(n_A+1)/2$, the smallest rank sum 8 sites could have ($1+2+\cdots+8=36$). The result is U.

Table 13: Ranking the pooled bat counts to calculate U.

Passes	Group	Rank
10	LED	1.0
11	LED	3.0
11	LED	3.0
11	LED	3.0
12	LED	5.0
16	Sodium	7.0
16	Sodium	7.0
16	Sodium	7.0
17	Sodium	10.0
17	Sodium	10.0
17	LED	10.0
18	LED	12.5

Passes	Group	Rank
18	LED	12.5
21	Sodium	14.0
22	Sodium	15.0
34	Sodium	16.0

The sodium ranks sum to $14 + 7 + 15 + 7 + 10 + 7 + 10 + 16 = 86$, so $U = 86 - 36 = 50$, matching the table R gave us above.

How the null distribution of U is built. Under H_0, both groups are draws from the same population. Once you pool and rank all $N = n_A + n_B = 16$ values, chance alone decides which ranks end up in the sodium group, so each of the $\binom{16}{8} = 12{,}870$ ways of splitting the 16 ranks into two groups of 8 is equally likely. ($\binom{16}{8}$, read "16 choose 8", counts the ways to pick 8 items out of 16.) No new sites are measured. Each split gives one U, the value our data would have produced if group labels had nothing to do with bat activity. Listing the U for all 12,870 splits gives the exact null distribution of U, shown in Figure 5.

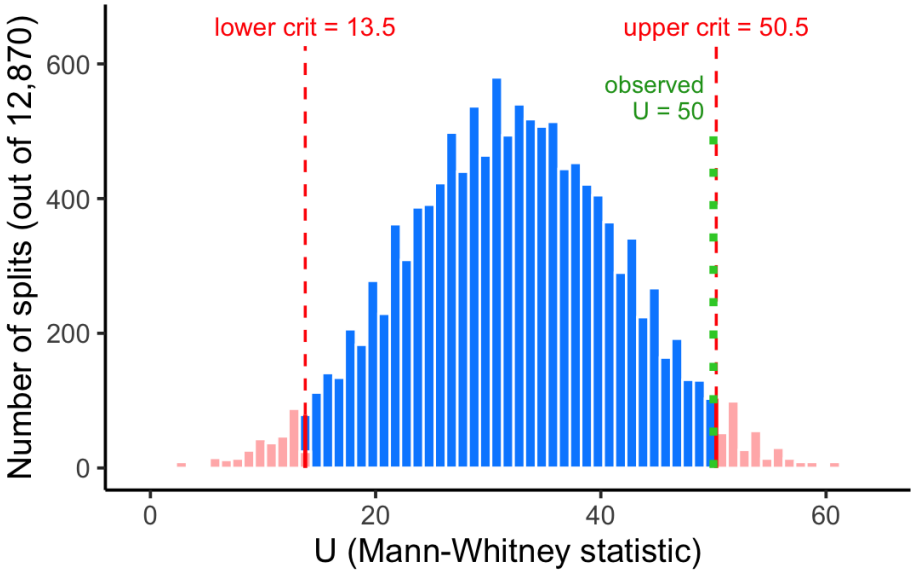


Figure 5: Exact null distribution of the Mann-Whitney U statistic, from all 12,870 equally likely ways to split the 16 pooled, ranked counts of bat passes into two groups of 8. Shaded regions mark the two-tailed 5% rejection region. Our observed U = 50 falls just short of the upper critical value, 50.5, outside the rejection region.

Our observed $U = 50$ falls just short of the upper critical value, 50.5, so it is not in the 5% region. A U of 50 or more comes from about 2.9% of the 12,870 splits, which puts the exact two-tailed p at 0.058. The result is not significant, though it is close to the cutoff. With tied counts, R uses a normal approximation to this distribution instead of the exact one, so its p-value, 0.063, differs slightly from the exact figure.

The test does not need normality because nothing in this distribution assumed a shape for bat activity. The only inputs were our 16 ranks and the rule from H_0 that every split is equally likely.

The full picture. The t-test gave $p = 0.024$, a significant result. The Mann-Whitney-Wilcoxon test on the same 16 sites gave $p = 0.063$, not significant at $\alpha = 0.05$. The t-test's p-value depends on a normality assumption that both groups fail, so it can't be taken at face value. A violated assumption can push a

p-value too low or too high, and the t-test alone gives no way to tell which. If the data had been roughly normal, the t-test would have been the better choice, because it has more power.

Table 14 collects the Mann-Whitney-Wilcoxon test on one card.

Table 14: Test card: Mann-Whitney-Wilcoxon test.

Purpose	Do two separate groups differ? Different units in each group, data not normal
Hypotheses	H_0: both groups have the same distribution (ranks well mixed between the groups). H_a: values tend to rank higher in one group than the other
Assumptions	Both groups' distributions have the same shape (normality not required)
Test statistic	Rank all values from both groups together. U = sum of Group A's ranks $- n_A(n_A + 1)/2$. R labels it W
Report	Direction, each group's median, U, p

Categorical data: the chi-square tests

Every test so far (the t-tests in Chapter 6, and the signed-rank and Mann-Whitney-Wilcoxon tests above) needs a continuous outcome variable. But a lot of environmental data isn't continuous at all: a site either has an invasive species or it doesn't; a stream sample either exceeds a safety threshold or it doesn't; a plot falls into one land-cover category, not a number line. Counts in categories need a different test, the chi-square (χ^2) test (Pearson 1900).

The **chi-square distribution** is the family of curves that the χ^2 statistic approximately follows when H_0 is true and the expected counts (the counts H_0 predicts for each category, defined below) are large enough. The test compares its statistic with this distribution. The symbol χ^2 is used for both the statistic and the distribution, the way t is. Because χ^2 adds up squared gaps between observed and expected counts, it can never be negative, and a mismatch in either direction makes it bigger. A χ^2 near 0 means the counts match the expectation closely, so only the upper tail counts against H_0. Like t, the chi-square distribution has one curve for each number of degrees of freedom (Figure 6). With few degrees of freedom it is sharply right-skewed, and as df grows it shifts right and becomes more symmetric.

Figure 7 shows what the rejection region looks like in practice, using the goodness-of-fit example we'll run below, which has $df = 3$. The critical value at $\alpha = 0.05$ is 7.81. When H_0 is true, a statistic falls beyond that point (the shaded region) 5% of the time.

For the rank tests, listing every possible arrangement of the ranks under H_0 gives the exact null distribution, though R used a normal approximation to it in both examples because of ties and zeros. For the chi-square tests, the chi-square curve approximates the null distribution of the statistic, and the approximation is good only when expected counts are large enough. Each chi-square test below therefore comes with an expected-count rule.

There are two chi-square tests, and the question decides which one to use. The goodness-of-fit test asks whether the counts for one categorical variable match an expected pattern. The test of independence asks whether two categorical variables are related.

The chi-square goodness-of-fit test

The **chi-square goodness-of-fit test** checks how well (the "goodness") the observed counts for one categorical variable match (the "fit") the counts expected under a stated pattern. The expectation can come from a historical baseline, a theory, or an assumption that every category is equally likely. A small chi-square statistic means a good fit, with observed counts close to expected. A large one means a poor fit, with observed counts far from expected.

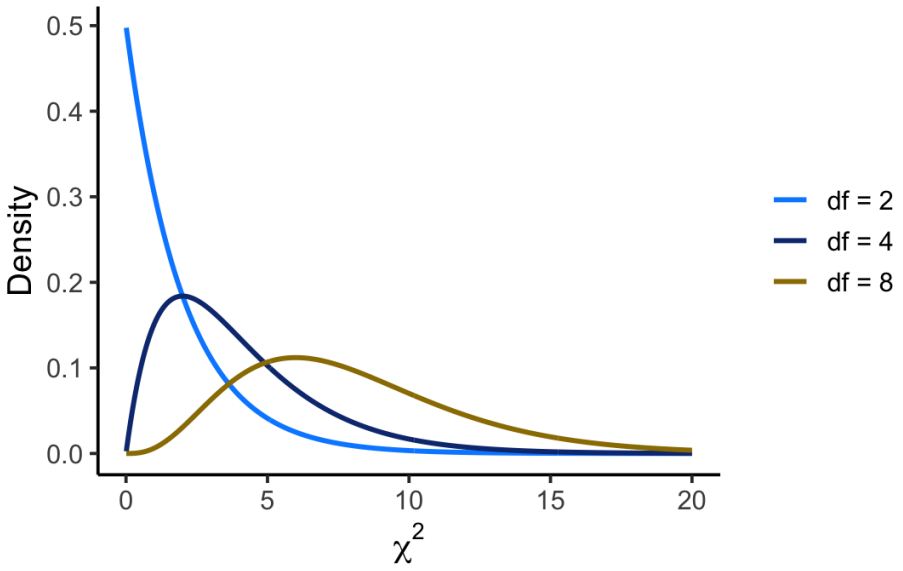


Figure 6: Chi-square distributions with 2, 4 and 8 degrees of freedom. Chi-square is never negative and is right-skewed. As df increases, the distribution shifts right and becomes more symmetric.

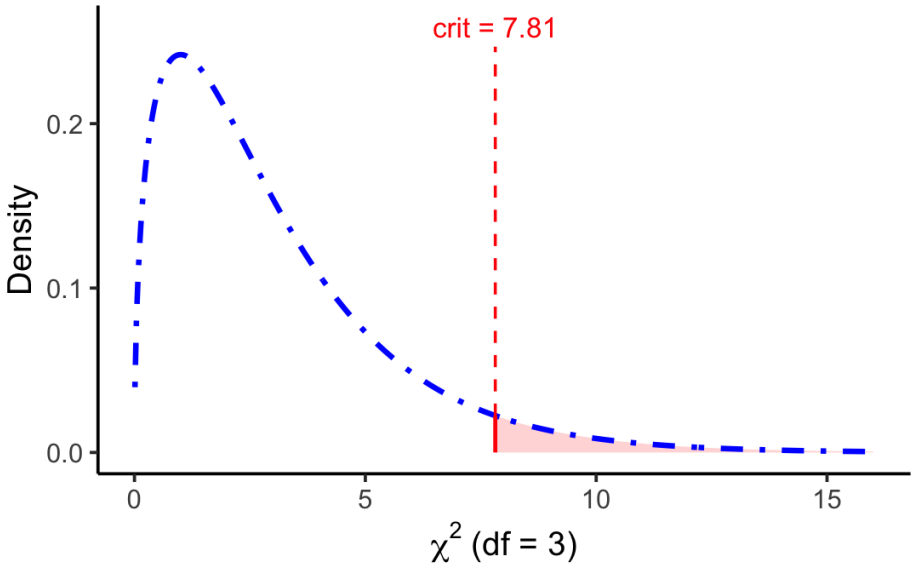


Figure 7: The chi-square distribution with df = 3. The shaded region marks the rejection region at $\alpha = 0.05$ (critical value 7.81).

A state wildlife agency has historically found fish evenly distributed across four zones of a lake, 25% in each. The counts in this example are invented. This year's survey of 200 fish found between 38 and 61 fish per zone. Figure 8 lays out the structure, which differs from every design so far in this book.

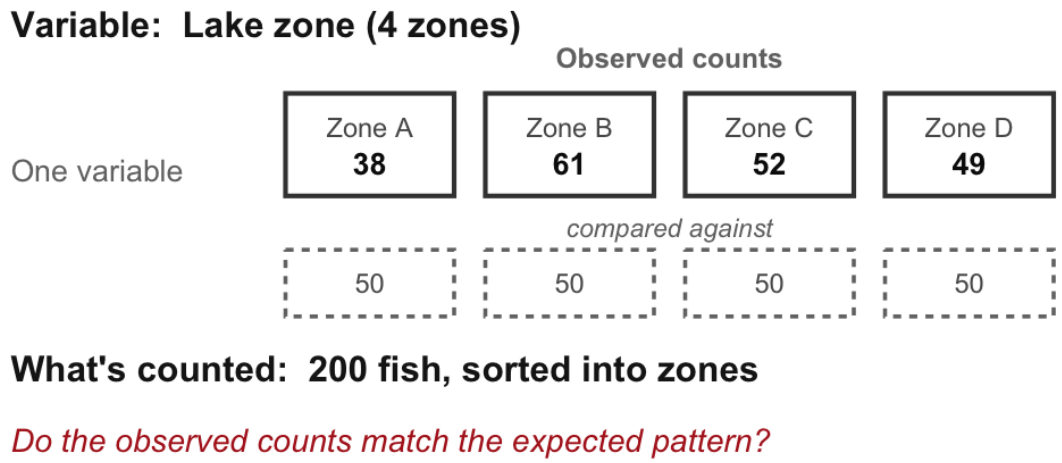


Figure 8: A goodness-of-fit design. The counts themselves are the data, with no dependent variable measured on each unit, and they are compared against an expected pattern rather than against another group.

Every other test so far has had a dependent variable measured on each unit, usually compared across groups that an independent variable defines. Here there is only one variable, lake zone, and nothing is measured on the fish beyond which zone they turned up in. The counts *are* the data, and the comparison is against an expectation rather than against another group.

Table 15: Observed fish counts by lake zone (n = 200).

Zone	Observed
A	38
B	61
C	52
D	49

The question. Does this year's distribution across zones differ from the historical 25%-each pattern, or is this the kind of unevenness you'd expect from random sampling alone?

- H_0: the proportions follow the stated pattern, 25% of fish in each zone, as in the historical baseline.
- H_a: the proportions do not follow that pattern.

```
chisq.test(observed_gof, p = rep(0.25, 4))
```

Table 16: Chi-square goodness-of-fit test.

Statistic (χ^2)	df	p
5.4	3	0.145

Assumption. The goodness-of-fit test makes no assumption about the shape of your data, since categories aren't numbers. It does require expected counts of about 5 or more in each category (Cochran 1954; Larntz 1978). The 5 is a conventional rule of thumb, not a value computed from these data or compared to a critical value. The rule tells you whether the chi-square curve is a trustworthy approximation. Check it before you run the test.

Here, expected count per zone is $200 \times 0.25 = 50$ (total fish times the 25%-per-zone expectation under H_0), comfortably above that minimum. The **expected count** for a category is the total count times the probability H_0 gives that category. This is the expected-value idea from Chapter 3 applied to counts. When expected counts are too small, a different method (like Fisher's exact test) is more appropriate.

χ^2 adds one term per category. For each category, subtract the expected count from the observed count, square the difference, and divide by the expected count. Then add these terms across categories: $\chi^2 = \sum \frac{(O-E)^2}{E}$.

Table 17: Chi-square contribution by zone.

Zone	Observed	Expected	Contribution
A	38	50	2.88
B	61	50	2.42
C	52	50	0.08
D	49	50	0.02

Summing the last column: $2.88 + 2.42 + 0.08 + 0.02 = 5.40$, matching R's statistic above. For goodness-of-fit, $df = k - 1$, where k is the number of categories. Once three zone counts are known, the fourth is fixed because all four must add to 200, so $df = 4 - 1 = 3$. With $df = 3$, the critical value at $\alpha = 0.05$ is 7.81 (the same one marked in Figure 7). Our 5.40 falls well short of it.

Interpretation. $\chi^2(3) = 5.40$, $p = 0.145$, so the result is not significant. This year's zone counts are consistent with the historical 25%-each pattern. The unevenness we see is within the range chance alone would produce.

Table 18 collects the chi-square goodness-of-fit test on one card.

Table 18: Test card: chi-square goodness-of-fit test.

Purpose	Do one categorical variable's counts match a pattern stated in advance?
Hypotheses	H_0: the proportions follow the stated pattern. H_a: they do not
Assumptions	Every expected count 5 or more (a pre-check, done before the test)
Test statistic	$\chi^2 = \sum(O-E)^2/E$, $E = n\times$ stated proportion, $df = k-1$ (k = number of categories)
Report	$\chi^2(df)$, p, and which categories came in above or below their expected counts

The chi-square test of independence

The **chi-square test of independence** asks whether two categorical variables measured on the same units are related. Suppose researchers classify 150 city blocks by land use and by whether each block's peak summer surface temperature, measured from satellite images, topped 40°C. The data in this example are invented. Every block falls into exactly one cell of Figure 9. A **cell** is one combination of categories in the table, here one land use with one temperature outcome.

In that table, nobody assigned anything. The blocks already had their land use and their summer temperatures before the researchers looked, and the cells count how many turned out each way. That makes this an

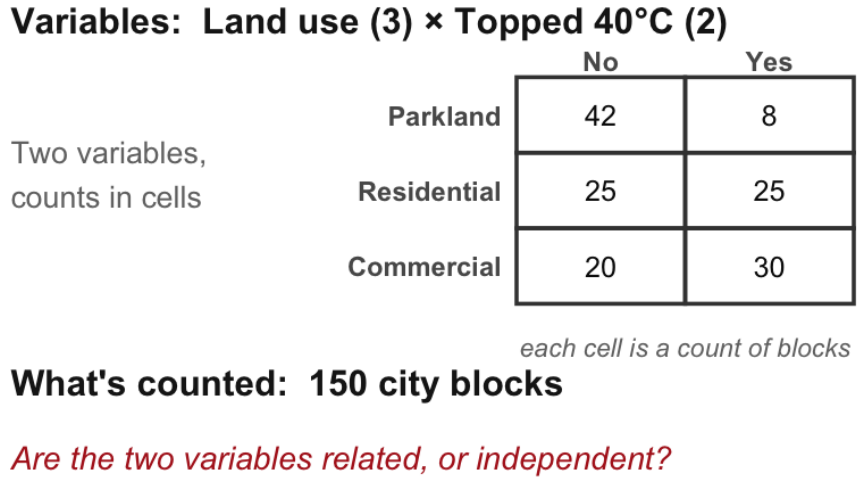


Figure 9: A test-of-independence design. Two categorical variables are crossed, and each cell holds the count of blocks that fell into that combination.

observational design. Each block's temperature has been reduced to a yes or no, so the cells hold counts of blocks rather than temperatures.

Table 19: City blocks by land use and whether summer surface temperature topped 40°C.

	No	Yes
Parkland	42	8
Residential	25	25
Commercial	20	30

The question. Is topping 40°C independent of land use, or are the two related?

- H_0: land use and topping 40°C are independent, so every land use has the same rate of blocks topping 40°C.
- H_a: land use and topping 40°C are related, so the rate differs between land uses.

```
chisq.test(tab)
```

Table 20: Chi-square test of independence.

Statistic (χ^2)	df	p
21.84	2	< 0.001

Assumption. Same rule as goodness-of-fit: expected counts should generally be 5 or more in every cell of the table. Unlike the goodness-of-fit test, where expected counts came from a stated proportion (25% per zone), the expected counts here come from the table's own row and column totals. In both chi-square tests, the expected count is the count H_0 predicts. In this test H_0 is independence. If land use and temperature are unrelated, parkland blocks should make up the same share of the 87 blocks that did not top 40°C as

they do of all 150 blocks, which is $\frac{50}{150}$, or one third. That gives an expected count of $87 \times \frac{50}{150} = 29$ for Parkland/No. In general, the expected count for a cell is $\frac{\text{row total} \times \text{column total}}{\text{grand total}}$.

Table 21: Expected counts under H0 (independence).

	No	Yes
Parkland	29	21
Residential	29	21
Commercial	29	21

All six expected counts are between 21 and 29, well above the 5-count minimum. This check does not tell us whether the test is significant. It only tells us whether the chi-square approximation is trustworthy here.

For a table, χ^2 adds one term per cell. The formula is the same as for goodness-of-fit, $\chi^2 = \sum \frac{(O-E)^2}{E}$, using the observed table and the expected table above.

Table 22: Chi-square contribution by cell.

	No	Yes
Parkland	5.83	8.05
Residential	0.55	0.76
Commercial	2.79	3.86

Summing all six cells: $5.83 + 8.05 + 0.55 + 0.76 + 2.79 + 3.86 = 21.84$, matching R's statistic above. For a test of independence, $df = (\text{rows} - 1)(\text{columns} - 1)$, here $(3-1)(2-1) = 2$. With the row and column totals fixed, filling in two cells of the No column determines every other cell. The number of blocks never enters. With $df = 2$, the critical value at $\alpha = 0.05$ is 5.99, and our 21.84 is far beyond it. The critical value at $\alpha = 0.001$ is 13.82, and 21.84 is beyond that too, so $p < 0.001$.

Interpretation. $\chi^2(2) = 21.84$, $p < 0.001$, so the result is significant. Topping 40°C is not independent of land use. To see how the two variables are related, look at the table. Divide within each row to get the share of blocks that topped 40°C for each land use (Table 23).

Table 23: Share of city blocks that topped 40°C, by land use.

Land use	Topped 40°C / total	Share
Parkland	8 / 50	16%
Residential	25 / 50	50%
Commercial	30 / 50	60%

Here every land use has 50 blocks, so the raw counts already compare fairly. With unequal row totals they would not, which is why we divide within each row. Residential and commercial blocks topped 40°C three to four times as often as parkland blocks. The test says the two variables are related, but, like correlation, it does not by itself establish that land use *causes* the heat.

Table 24 collects the chi-square test of independence on one card.

Table 24: Test card: chi-square test of independence.

Purpose	Are two categorical variables, measured on the same units, related?
Hypotheses	H_0: the variables are independent (every row has the same rate). H_a: they are related
Assumptions	Every expected count 5 or more (a pre-check, done before the test)
Test statistic	$\chi^2 = \sum(O-E)^2/E$, $E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$, $df = (r-1)(c-1)$ (r rows, c columns)
Report	$\chi^2(df)$, p, and the rate within each row (which category drives it)

Table 25 collects the tests in this chapter by the kind of data and design each one handles.

Table 25: Choosing a test in this chapter.

Outcome	Design	Parametric test (Chapter 6)	Test in this chapter
Continuous	One group vs. a reference, or paired (before/after)	One-sample or paired t-test	Wilcoxon signed-rank test; sign test if the differences are not symmetric
Continuous	Two independent groups	Independent-samples t-test	Mann-Whitney-Wilcoxon test
Ordinal	Either design above	None	The same rank tests
Categorical (counts)	One categorical variable against an expected pattern	None	Chi-square goodness-of-fit test
Categorical (counts)	Two categorical variables	None	Chi-square test of independence

! Results reporting: nonparametric and chi-square tests

These tests report differently from a t-test. There is no mean difference to quote, because none of them used the means. Give the test's own statistic, the p-value, and, for the rank tests, a median rather than a mean.
Wilcoxon signed-rank, on the restoration differences:

> Soil moisture did not change significantly after restoration, Wilcoxon signed-rank $V = 39$, $p = 0.057$ (median change +1.5 percentage points across all 12 plots; 9 nonzero differences).

Mann-Whitney-Wilcoxon, on the bat activity comparison:

> Bat activity did not differ significantly between sites under sodium and LED streetlights, $U = 50$, $p = 0.063$ (medians 17 vs. 11.5 passes per night).

Chi-square, where the counts themselves are the result:

> Whether a city block topped 40°C was not independent of land use, $\chi^2(2) = 21.84$, $p < 0.001$. Parkland blocks topped 40°C in 8 of 50 cases, compared with 25 of 50 residential and 30 of 50 commercial blocks.

Report the number of nonzero differences for a signed-rank test, since dropped zeros change the effective sample size. For chi-square, give the counts as well as the statistic, because the counts are what a reader can interpret.

Chapter Summary

Why it matters. Environmental data often fail the assumptions a t-test needs. Samples can be small, distributions can be skewed, and some outcomes are counts in categories instead of measurements. This chapter covers the tests for those cases and how to tell when you need one. Chapter 8 compares three or more groups, with the Kruskal-Wallis test as its rank-based option.

Core ideas

- **Rank-based tests trade power for weaker assumptions.** The Wilcoxon signed-rank and Mann-Whitney-Wilcoxon tests ask the same questions as the one-sample or paired and the independent-samples t-tests, but work from ranks instead of raw magnitudes.
- **The sign test uses even less of the data.** It looks only at the direction of each difference. On the same restoration data, the t-test, signed-rank test, and sign test gave p-values of 0.026, 0.057 and 0.508, because each one uses less of the data than the last.
- **Nonparametric tests have their own assumptions.** The signed-rank test assumes the differences are symmetric about their median. Mann-Whitney-Wilcoxon assumes the two distributions have the same *shape*, which is not the same as equal variance. State the assumption you are actually making.
- **Check assumptions before you run a test.** Choosing the test after seeing its result lets the result decide the test. Both worked examples ran the parametric test first only to show how its p-value can mislead when an assumption fails.
- **Categorical data needs a different tool entirely.** The chi-square tests handle counts: goodness-of-fit asks whether one variable's counts match an expected pattern, and the test of independence asks whether two categorical variables are related.
- **The testing logic never changes.** State H_0 and H_a, compute a statistic, and ask how surprising that statistic would be if H_0 were true. Only the statistic and its null distribution differ from test to test.

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